

Use the AnnotationHub package to obtain data on "CpG Islands" in the human genome.
How many islands exist on the autosomes?

26567

26641

24696

25495

How many CpG Islands exist on chromosome 4?

1004

1031

1019

1011

Obtain the data for the H3K4me3 histone modification for the H1 cell line from
Epigenomics Roadmap, using AnnotationHub. Subset these regions to only keep regions
mapped to the autosomes (chromosomes 1 to 22). How many bases does these regions
cover?

42600796

41135164

37650561

38610551

Obtain the data for the H3K27me3 histone modification for the H1 cell line from Epigenomics Roadmap, using the AnnotationHub package. Subset these regions to only keep regions mapped to the autosomes. In the return data, each region has an associated "signalValue". What is the mean signalValue across all regions on the standard chromosomes?

4.799766

4.372933

4.770728

5.140497

Bivalent regions are bound by both H3K4me3 and H3K27me3. Using the regions we have obtained above, how many bases on the standard chromosomes are bivalently marked?

10289096

10207246

9926503

10984729

We will examine the extent to which bivalent regions overlap CpG Islands. How big a fraction (expressed as a number between 0 and 1) of the bivalent regions, overlap one or more CpG Islands?

0.5059474

0.5140894

0.5383644

0.5839835

How big a fraction (expressed as a number between 0 and 1) of the bases, which are part of CpG Islands, are also bivalent marked?

0.1813049

0.241688

0.1756357

0.2299431

How many bases are bivalently marked within 10kb of CpG Islands? **Tip:** consider using the "resize()" function.

9782086

10293363

9189910

7433923

How big a fraction (expressed as a number between 0 and 1) of the human genome is contained in a CpG Island?

Tip 1: the object returned by AnnotationHub contains "seqlengths".

Tip 2: you may encounter an integer overflow. As described in the session on R Basic Types, you can address this by converting integers to numeric before summing them, "as.numeric()".

0.007180816

0.007047481

0.005185928

0.006040689

Compute an odds-ratio for the overlap of bivalent marks with CpG islands.

138.4391

169.0962

156.0433

160.4022